

When Simple Wins: Lightweight CNN Encoders for Resource-Constrained Nucleus Segmentation

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Take-home message

In a controlled comparison with only the encoder varied, a VGG U-Net trained from scratch matched ResNet and outperformed an ImageNet-pretrained Swin-T on PanNuke and MoNuSeg, at lower inference latency and memory.

Training data
37 MoNuSeg slides

Hardware
one GPU; fits in 8 GB

Pretraining
none (VGG)

Inference
7.1 ms vs 18.4 ms (Swin-T)

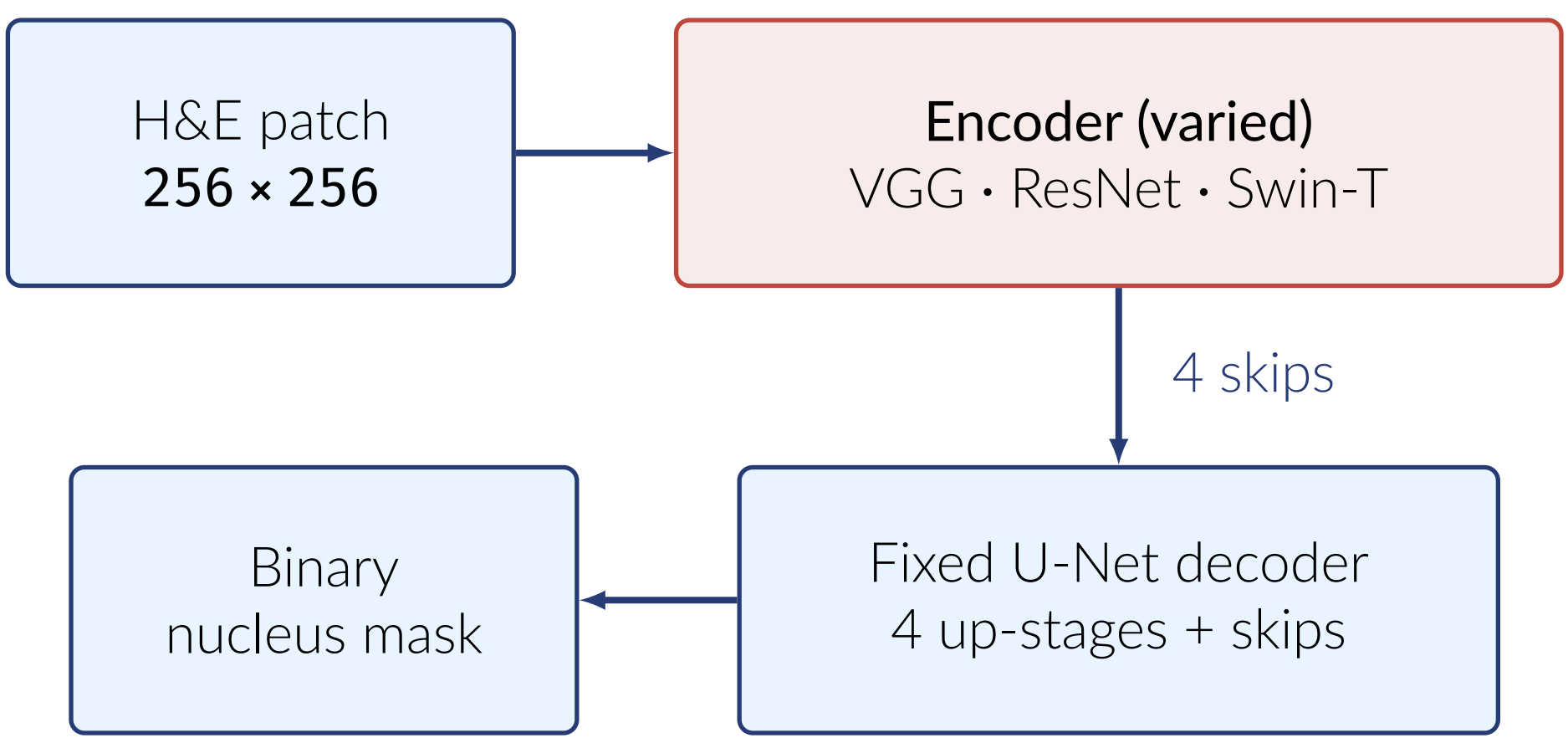
Motivation

Pathology foundation models (UNI, 307M parameters; Virchow2, 632M) ease the labelled-data shortage, but they are hard to fine-tune, questioned under cross-site shift, and transfer poorly beyond H&E. Edge microscopy, veterinary and rare-disease labs often have **only tens of annotated slides and a single GPU**.

Question

Among lightweight encoders that can be trained from scratch, which is best in the low-data regime?

Controlled design: only the encoder changes



The decoder, data pipeline, augmentation, optimiser and evaluation are shared, and every encoder returns four feature maps through one interface. Results are means over **5 training seeds** on one V100, with paired 95% CIs.

Encoder	First skip	Params
VGG (plain 3×3 stacks)	256^2	34.0M
ResNet (residual blocks)	256^2	37.5M
Swin-T (scratch / ImageNet)	$64^2 \rightarrow$ upsampled	86.8M

Datasets

PanNuke

19 tissue types, 256×256 patches. We train on fold1 (2,656 patches), validate on fold2 and test on fold3.

MoNuSeg

37 training slides (1000×1000 , 7 organs) and 14 test slides, including organs never seen in training. Slides are tiled into 256×256 patches and split by slide.

PanNuke: full results (fold3, 5 seeds)

Encoder	Loss	Dice	IoU	Epochs	Params
Otsu (classical)	—	0.482	0.349	—	0
Swin-T (scratch)	BCE	0.825 (.001)	0.703 (.002)	69.0 (9.4)	86.8M
Swin-T (pretrained)	BCE	0.824 (.001)	0.701 (.001)	96.2 (5.0)	86.8M
ResNet	BCE	0.853 (.000)	0.743 (.001)	49.8 (3.8)	37.5M
VGG	Dice	0.856 (.001)	0.748 (.001)	84.8 (17.5)	34.0M
VGG	BCE	0.854 (.000)	0.746 (.000)	51.8 (2.6)	34.0M

Mean (standard deviation) over 5 training seeds; Otsu is deterministic. Epochs is the early-stopping epoch. Bold marks the column maximum; VGG and ResNet are not ranked.

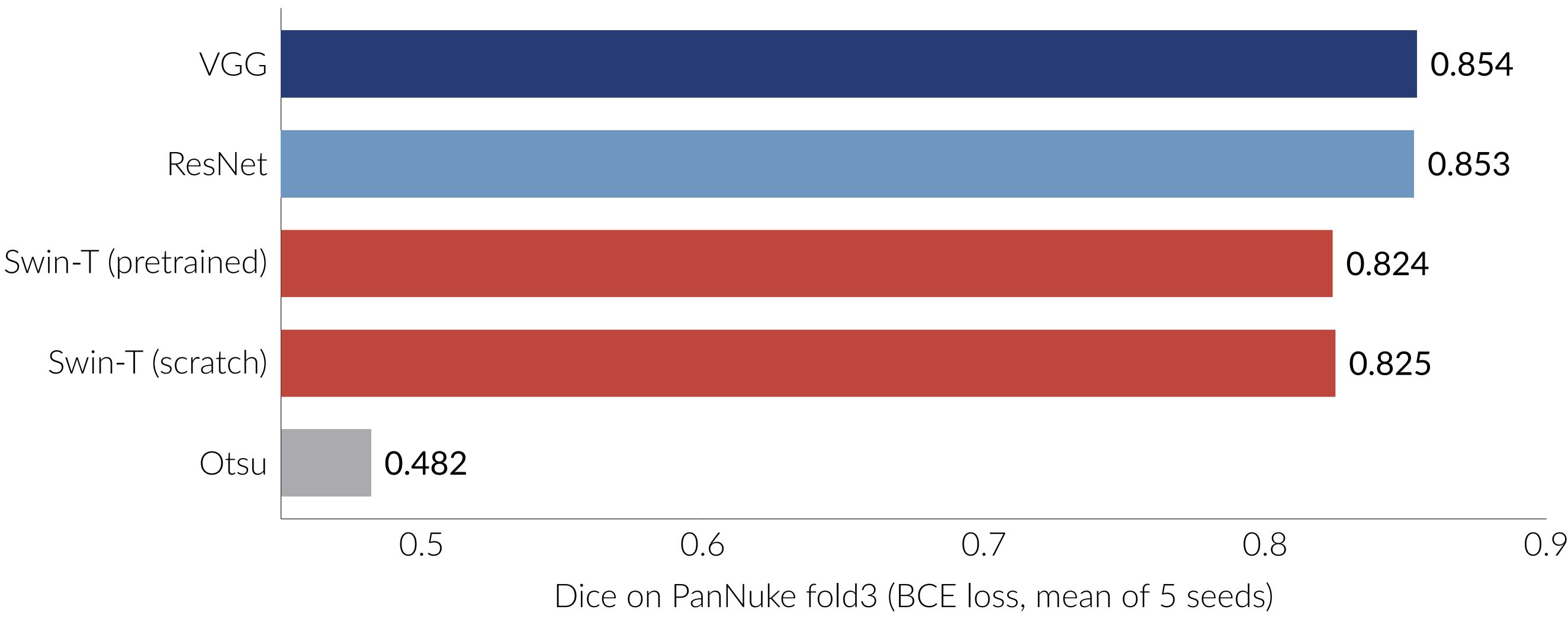
Why should CNNs win here?

- Skip resolution.** Nucleus boundaries are a few pixels wide. VGG keeps full resolution at its first skip; Swin's 4×4 tokens downsample $4\times$ immediately, and upsampling can't recover the detail.
- Inductive bias vs data.** Attention has to learn the locality that convolution builds in, and a few thousand patches isn't enough data for that, even with ImageNet weights.
- Input scale.** On 256^2 patches, VGG's receptive field already covers most of the input by stage 4.

Limitations

- 5 seeds on one fixed split, so split variance isn't bounded.
- No learning-rate sweep, and the LR schedule differs between CNN and Swin.
- Only three encoders. At 34M parameters VGG isn't a truly compact model.
- Binary masks only, not instance separation. H&E stains only.

PanNuke: VGG $\approx$ ResNet $>$ Swin



- VGG vs ResNet:** 0.002 Dice (CI [0.001, 0.002]), below the 0.005 we treat as meaningful, so we don't rank them. Residual shortcuts don't help at four stages.
- VGG vs pretrained Swin:** 0.030 Dice (CI [0.029, 0.031]).
- Pretraining** moves Swin by only 0.001, so the gap doesn't come from the ImageNet weights.
- BCE vs Dice loss:** 0.854 vs 0.856, which is immaterial.

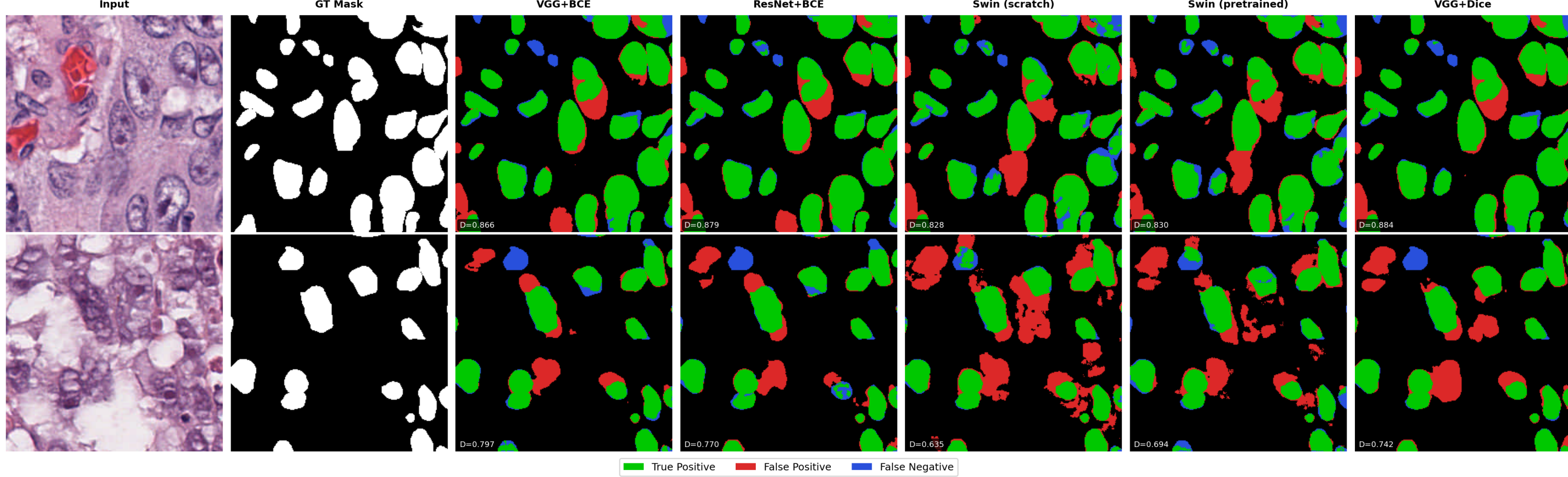


Figure 1. Two PanNuke test patches, one seed. The lower patch was picked because the gap is large (VGG 0.797, pretrained Swin 0.694); in the upper one ResNet and VGG+Dice edge out VGG+BCE. Swin's errors are mostly false-positive blobs on background tissue.

MoNuSeg: 37 training images

Model	Dice	IoU	Params
MedT (published, 30 images) [†]	0.796	0.662	1.4M
Swin-T pretrained (ours)	0.758 (.007)	0.613 (.008)	86.8M
VGG, from scratch (ours)	0.794 (.011)	0.660 (.015)	34.0M

VGG beats pretrained Swin by **0.036 Dice** (CI [0.021, 0.052]), a gap similar to the one on PanNuke.

[†]A cross-paper anchor rather than a controlled comparison: MedT used 30 training images and 512^2 input. We read it only as VGG reaching MedT's level.

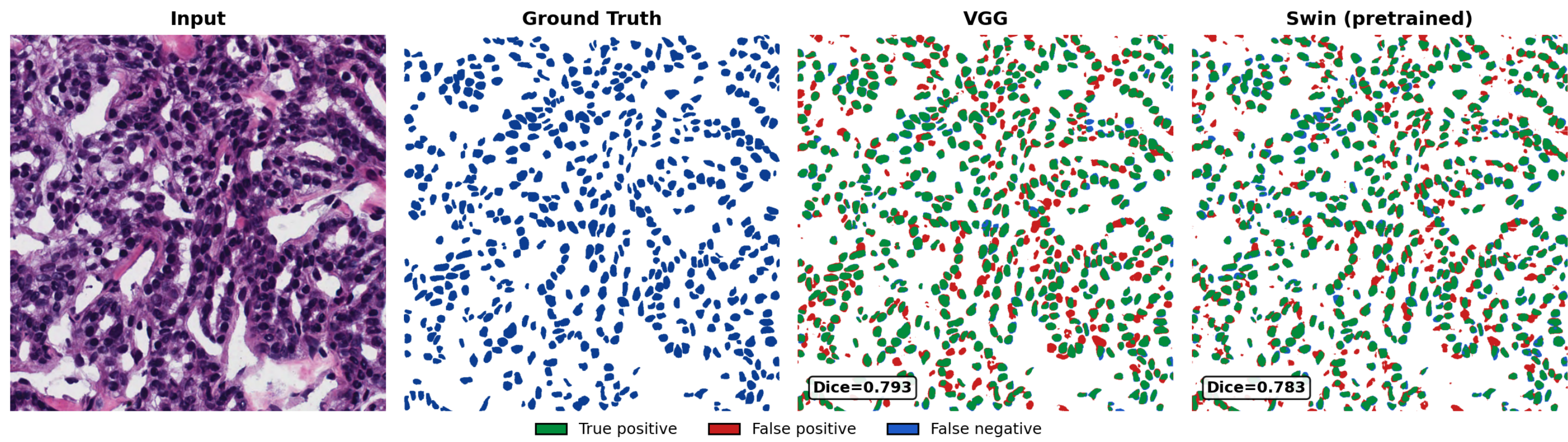


Figure 2. A dense MoNuSeg test region, chosen for how dense it is rather than for the size of the gap (0.010 Dice here vs 0.036 on average). Swin merges neighbouring nuclei across narrow gaps that VGG keeps separate.

Inference cost (256^2 , batch 1, fp32, V100)

Model	Params	Latency	Peak memory
VGG U-Net	34.0M	7.1 ms	265 MB
ResNet U-Net	37.5M	9.6 ms	277 MB
Swin-T U-Net	86.8M	18.4 ms	619 MB

VGG beats Swin-T on accuracy, latency and memory. Every model we trained fits in 8 GB (peak training memory: 3.6 GiB for VGG, 5.4 GiB for Swin).

Next steps

A resolution-matched Swin to separate tokenisation from encoder family; H&E-matched pretraining (e.g. UNI); the data scale at which Swin catches up; compact encoders (MobileNet, ConvNeXt-T); non-H&E stains.